Interview Question

1. Tell me about yourself?

2: describe current project?

Core Java

1. What is Immutable class and How to create an immutable class(Immutable)?
2. Can we extends String class? Ans- No because it is final class
3. Can we extends Static class?
Ans. extending static classes is allowed, since its members are not necessarily static. the static modifier can only be used on nested classes because it can only be used on class members (and only nested classes can be class members).
Q. Difference b/w Abstract Class and Interface.
Q. What is class and interfaces in java?
Q. Difference b/w String & string buffer, String Builder?

Q. Internal work of ENUM

Q. Can we extends class in enum?

Q. How to Handle OutOfMemoryError Exceptions in Java?

Ans: A java.lang.OutOfMemoryError is a runtime error in Java which occurs when the Java Virtual Machine (JVM) is unable to allocate an object due to insufficient space in the Java heap. The Java Garbage Collector (GC) cannot free up the space required for a new object, which causes a java.lang.OutOfMemoryError. This error can also be thrown when the native memory is insufficient to support the loading of a Java class.

Ask about Java 8 Features:

- Lambda expressions,
- Method references,
- Functional interfaces,
- Stream API,
- Default methods,
- Base64 Encode Decode,
- Static methods in interface,
- Optional class,
- Collectors class,

- ForEach() method,
- Parallel array sorting

Ques: Java 8 Stream : Differences b/w Intermediate And Terminal Operations?

Ans: 1) The main difference between intermediate and terminal operations is that intermediate operations return a stream as a result and terminal operations return non-stream values like primitive or object or collection or may not return anything.
2) As intermediate operations return another stream as a result, they can be chained together to form a pipeline of operations. Terminal operations can not be chained together.
3) Pipeline of operations may contain any number of intermediate operations, but there has to be only one terminal operation, that too at the end of pipeline.
4) Intermediate operations are lazily loaded. When you call intermediate operations, they are actually not executed. They are just stored in the memory and executed when the terminal operation is called on the stream.
5) As the names suggest, intermediate operations doesn't give end result. They just transform one stream to another stream. On the other hand, terminal operations give end result.

6) Intermediate Operations :

map(), filter(), distinct(), sorted(), limit(), skip()

Q: Difference Between Collections And Streams In Java?

1. **Collections and Streams, both are conceptually two different things which are used for two different purposes. If the collections are used to store the data then the streams are used to perform operations on that data.**
2. **Collections are used to store and group the data in a particular data structure like List, Set or Map. But, streams are used to perform complex data processing operations like filtering, matching, mapping etc on stored data such as arrays, collections or I/O resources.**
3. **You can add to or remove elements from collections. But, you can't add to or remove elements from streams. Stream consumes a source, performs operations on it and returns a result. They don't modify even the source also.**
   External Iteration Vs Internal Iteration
4. **The main specialty of Java 8 Streams is that you need not to worry about iteration while using streams. Streams perform iteration internally behind the scene for you. You just have to mention the operations to be performed on a source.**

5. **Streams are traversable only once. If you traverse the stream once, it is said to be consumed. To traverse it again, you have to get new stream from the source again. But, collections can be traversed multiple times.**

collection

Q.What is the difference between Comparable and Comparator?
Q.Difference between hashcode() and equals methods?

Java.lang.object has two very important methods defined:

- **public boolean equals(Object obj) and**
- **public int hashCode().**

Q. What is the difference between Array and ArrayList?
Q. Difference between Hashmap and Concurrent Hashmap?
Q. why hashmap contains only single null key?
Q. Internal working of Hashmap?
Q. Internal working of Treeset?
Q. Internal working of Hashset?

Design Pattern

Q. Singleton Design Pattern
Q. Factory Design
Q. And Abstact Factory Pattern
Q. Differences b/w Factory Design And Abstact Factory Pattern

Multithreding
Ques: Race Condition?
Q. What is synchronization and types of implementation?

Q. Can we call wait method out side synchronized blocked?

Q. Interrupted thread?

Spring & Spring Boot

Q. Scope of bean?
Q. What is DI & IOC.
Q. How to activate logging in spring boot and how many types of LOGGING levels?
Q. Spring boot annotations?

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Hibernate

Q: how to create composite key in Hibernate?
Ans: By @EmbeddedId,@Embeddable
Q. What is JPA?
Q. What is hibernate?
Q. Difference b\w Hibernate and JPA?

Q. List the key components of Hibernate.

Key components of Hibernate are:

- Configuration
- Session
- SessionFactory
- Criteria
- Query
- Transaction

Microservice

1. **What is microservices and advantages and disadvantages?**

2. **Microservice Design Pattern architecture**

    - **API Gateway Design Pattern**
    - **Circuit Breaker Pattern**
        - **Hypothtical pattern**
        - **Saga Design pattern**
3. **Which Microservice design patteren have you used in your project? Explain your microservice architecture?**
4. **What is Saga Design pattern**
5. **Components of Microservices**

There are the following components of microservices:

- Spring Cloud Config Server

- Netflix Eureka Naming Server
- Hystrix Server
- Netflix ZuulAPI Gateway Server
- Netflix Ribbon
- Zipkin Distributed Tracing Server

Q. How to Check logs in Kubernetes?

Database:

Q: Third highest salary of an Employee department wise?

Program :

Q: Find missing values from array

Arr [] = {1,3,5,9}
Output: 7

Q: find values from array that equals to given Sum

```
int [] arr = {2,7,11,15};
int total = 9;

Output: {2,7}
```

Q: Reverse String and Reverse Paragraph.

TCS

Ques: How Hash Collisions Are Handled in Java 8?

Ans:

1. **Traditional Approach (Java 7 and earlier)**

- Each bucket in a HashMap stored entries in a **linked list**.
- If many keys hashed to the same bucket, lookup time degraded from **O(1)** to **O(n)**.
- Worst case: all keys in one bucket → very slow performance.

2. **Java 8 Enhancement**

- **Linked List → Red-Black Tree conversion**:
  - If a bucket's size exceeds **TREEIFY_THRESHOLD (8)**, the linked list is converted into a **Red-Black Tree**.
  - This improves lookup time from **O(n)** to **O(log n)**.
- **Capacity Check**:
  - Conversion happens only if the overall table capacity is at least **MIN_TREEIFY_CAPACITY (64)**.
  - If capacity is smaller, the map resizes instead of treeifying.
- **Untreeify**:
- If entries in a tree bucket shrink below **UNTREEIFY_THRESHOLD (6)**, it converts back to a linked list.

3. **Bit Spreading in Hash Function**

- Java 8 introduced a new hash() method:

```
static final int hash(Object key) {
   int h;
  return (key == null) ? 0 : (h = key.hashCode()) ^ (h >>> 16);
}
```

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